

Highlights

Somatic and affective symptom clusters among reproductive-aged women in Bangladesh: a network-informed latent profile analysis

- Network-Informed LPA reveals distinct symptom phenotypes beyond simple severity.
- Silent Somatic profile identifies hidden distress in 49.4% of the sample.
- Fatigue acts as the central bridge between somatic and affective symptom clusters.
- Higher education and wealth paradoxically predict severe comorbid profile membership.

Binary diagnoses miss complex somatic-affective patterns in reproductive-aged women.

Somatic and affective symptom clusters among reproductive-aged women in Bangladesh: a network-informed latent profile analysis

Abstract

Most mental health studies in Bangladesh are based on variable-centered methods and binary classification of individuals. Though common in lower and lower-middle-income countries (LLMICs), it still remains under-researched in public health literature. . This study utilized a network-based person-centered approach to address these gaps by using a network-based and person-centered approach to identify the distinct psychopathological phenotypes and their sociodemographic determinants. Data were taken from the Bangladesh Demographic and Health Survey (BDHS-2022). Latent Profile analysis (LPA) was applied to depression and anxiety symptoms to identify non-overlap subgroups. Psychometric Network Analysis (PNA) was then employed to determine the center nodes of the symptoms. Finally, multinomial Logistic Regression was applied to the sociodemographic variables. Three mutually exclusive profiles, such as Resilient (25.6%), Silent Somatic (49.4%), and Severe Comorbid (25.0%), were created by LPA. Whereas, Network analysis figured out that Fatigue (strength = 1.958096) is the most central node. Finally, multinomial regression points out a paradoxical vulnerability of reproductive-aged women with higher education, and high SES had higher odds of belonging to the severe comorbid profile compared to the resilient profile. In this study, it is found that women with sadness, worry, and fear showed lower levels of depression and anxiety than fatigue and sleep. Here, population is considered heterogeneous and characterized by unique subgroups. Moreover, women possess high socioeconomic status and higher education are at significant risk of being classified in the severe comorbid class.

Keywords: Latent Profile Analysis, Psychometric Network Analysis,

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1. Research in Context

Evidence before this study: Associate information was accumulated before conducting this research. Several sources like PubMed, Google Scholar, Science Direct, and so on were utilized to know the about the existing works and their research gaps. Different types of keywords such as ‘somatic symptoms’, ‘network analysis’, ‘Latent profile analysis’, ‘prevalence of depression and anxiety in Bangladesh’, and ‘Indian subcontinent’ used to find related information about mental health research. After conducting rigorous search, it is found that most studies in Bangladesh rely on variable-centered approaches of depression and anxiety. Binary classification cut-off (e.g., ‘depressed’, vs ‘not depressed’) widely used which may ignores sub-clinical phenotypes, and solely sometimes statistical association between mental health symptoms and other variables presents as outcome.

Added value of this study: This study tries to address these gaps, and applied methods different from typical research. This study among the very few studies integrated network analysis with latent profile analysis to identify mental health phenotypes and applied on nationally representative sample (BDHS 2022). Later, multinomial logistics regression employed on sociodemographic data. This study creates three mutually exclusive profiles, among them 49.4% sample fell into a ‘silent somatic’ profile. Where network analysis figures out ‘fatigue’ as the central nodes, may contributing the most into severe psychological comorbidity.

Implications of all the available evidence: After analyzing all the available evidence and findings of this study suggest that mental health screening should move beyond traditional mood-based assessments. Where much huge quantity of sample in ‘silent somatic’ profile indicates many previous studies may falsely classified. It is also found that women with high SES and education are more tends to suffer mental health which is explained by "Urbanization stress" and "Role Conflict Theory".

2. Introduction

Depression and Anxiety are the major sources of mental health disorders that influence a person’s productivity, family bondings, and how to deal with the stresses of life. On the other hand, good mental health is recognized as a

basic human right that is shaped by several factors, like social, cultural, and political [1]. More than a billion people are living with mental health issues [1]. In Bangladesh, people of different groups, such as teenagers, university students, adolescents, reproductive women, and especially females [2, 3], are more vulnerable to mental health disorders. Several studies reported that the prevalence of mild to severe symptoms of depression and anxiety varied from 25% to around 90% [2, 3], such as 47.8% prevalence of depression in the 15-19-year-olds group [4], up to 71% for university students [5].

However, most of the existing studies in Bangladesh highlighted on variable-centered [6] approach and binary classification [7]. PHQ-9 and GAD-7 scales are used to classify whether they have a mental disorder or not; for instance, applying an arbitrary cut-off PHQ-9 score (score > 10) to classify as depressed and not depressed, and vice versa [7]. Moreover, in a variable-centered approach, every individual is assumed to have a similar relationship between symptoms [8]. Apart from this, they assume that two individuals with the same score are classified in the same manner without considering their socioeconomic factors [9]. These studies may mislead the conclusion since this score is calculated on the basis of the sum of an individual’s symptoms, because an individual does not have a problem with all the anxiety and depression symptoms at once.

Furthermore, traditional statistical models (regression, logistic regression) often identify poor education, low income, living in an urban or rural area, these single factors as potential risk of having mental health disorders. This ignores the potential “vulnerability paradox,” like richer or richest people may face social isolation, lack of security, and more working hours, which could lead to mental disorders, or people living in rural areas with low facilities may have more symptoms of depression and anxiety. Apart from these, several works have been carried out to predict mental health disorders by applying traditional machine learning and deep learning models. Models such as Random Forest, Support Vector Machines, genetic algorithm-optimized logistic regression, hybrid stacking, and deep neural networks showed efficient classification (often over 90% accuracy) capability to predict depression, anxiety, mental disorder, and suggest treatments [10]. Recently, to improve the model’s transparency and interpretability, explainable artificial intelligence (XAI) techniques have been introduced, which takes the prediction accuracy to 99% [11]. However, this model is still based on binary classification.

This present study addresses these drawbacks using a nationally represented dataset of women at their reproductive age and associated socio-

demographic data from the Bangladesh Demographic and Health Survey (BDHS) 2022. To facilitate a more advanced understanding of depression and anxiety in women in Bangladesh, this study (1) utilized Latent Profile Analysis (LPA) to identify unique latent profiles of depression and anxiety symptoms, (2) applied Psychometric Network Analysis (PNA) to find driver nodes or central symptoms that trigger the other symptoms of mental issues, (3) used Multinomial Logistic Regression (MNLogit) to examine sociodemographic variables and draw a probability based conclusion. By integrating profile and network-based this study goes beyond binary classification and regular measures of association of variables with mental health disorders. This study does not focus on how many women or what percentage of people have mental health issues; instead, it aims to examine the whole network of symptoms and how they connect.

2.1. Background Study

2.1.1. Statistical Analysis

Recent findings from the 2022 Bangladesh Demographic and Health Survey (BDHS) show a significant mental health burden among women of reproductive age; national prevalence rates for anxiety and depression were 19.5% and 4.9%, respectively [12]. Regional variations were evident, as women from Rangpur and Khulna faced higher odds of both conditions [12]. Furthermore, non-Muslim women were less likely to experience these disorders than Muslim women [12, 13].

Only 20.8% of impacted women reported seeking assistance, which is startlingly low given the rising prevalence [14]. Higher education, employment, and internet use were all positively correlated with seeking assistance, while housewives and smartphone users were less likely to do so [14, 15]. In certain SWPER index domains, empowerment paradoxically demonstrated an inverse relationship with seeking assistance, indicating sociocultural barriers to formal care utilization [15].

Awareness of mental health conditions (MHCs) is still extremely low, with only 8.5% and 6.2% of respondents aware of depression and anxiety, respectively, and more than half unaware of any MHC [16]. Education and financial stability were found to be consistently predictive of increased awareness [16].

Adolescents and university students have significantly higher prevalence estimates, ranging from 18% to 61% for anxiety and 26% to 74% for depression [17]. Poor sleep quality, physical inactivity, overuse of the internet, family

conflict, and academic stress are all significant correlates. Gender differences persist, with females being significantly more vulnerable [18, 17].

According to research, Bangladesh faces three challenges: persistently high socio-cultural barriers, critically low awareness and help-seeking rates, and high and regionally clustered prevalence. Bangladesh’s mental health crisis requires integrated, gender-sensitive interventions that address regional disparities, education, family dynamics, and digital literacy.

2.1.2. Predictive Analysis

Recent studies have increasingly employed machine learning (ML) to predict mental health outcomes, such as stress, anxiety, and depression, in Bangladeshi populations. The reported prevalence of stress (32%), anxiety (69–83%), and depression (56–85%) varies widely, especially among college students [19, 20]. Important correlates include academic discipline, smoking, social media addiction, hostile family dynamics, female gender, and sleep distress [19, 20].

Predictive performance differed by dataset and outcome across ML models. While Random Forest (RF) and XGBoost consistently performed well for depression and stress classification (accuracy $\approx$ 0.89–0.90) [21, 22], Support Vector Machine (SVM) showed superior accuracy for anxiety prediction (up to 91.5%) [19, 21]. Furthermore, competitive accuracy of over 90% was attained by CatBoost and KNN in certain studies [19, 22]. In feature importance analyses, it was often discovered that family ties, academic year, faculty of study, and internet/social media use were dominant predictors [19, 20].

CNNs are 96-97% accurate in predicting anxiety and depression [23]. However, issues with class imbalance and moderate ROC-AUC scores indicate limitations in generalizability [19, 24].

All things considered, ML-based predictive modeling has a lot of potential for early psychological distress detection, especially in young people. These models may be more useful for data-driven, scalable mental health screening in low-resource environments if socio-behavioral and digital indicators are incorporated.

3. Methodology

3.1. Data Source and Participants

This study utilized Bangladesh Demographic and Health Survey (BDHS) data from the demographic and health surveys (DHS) program in 2022. It is

Methodology Diagram

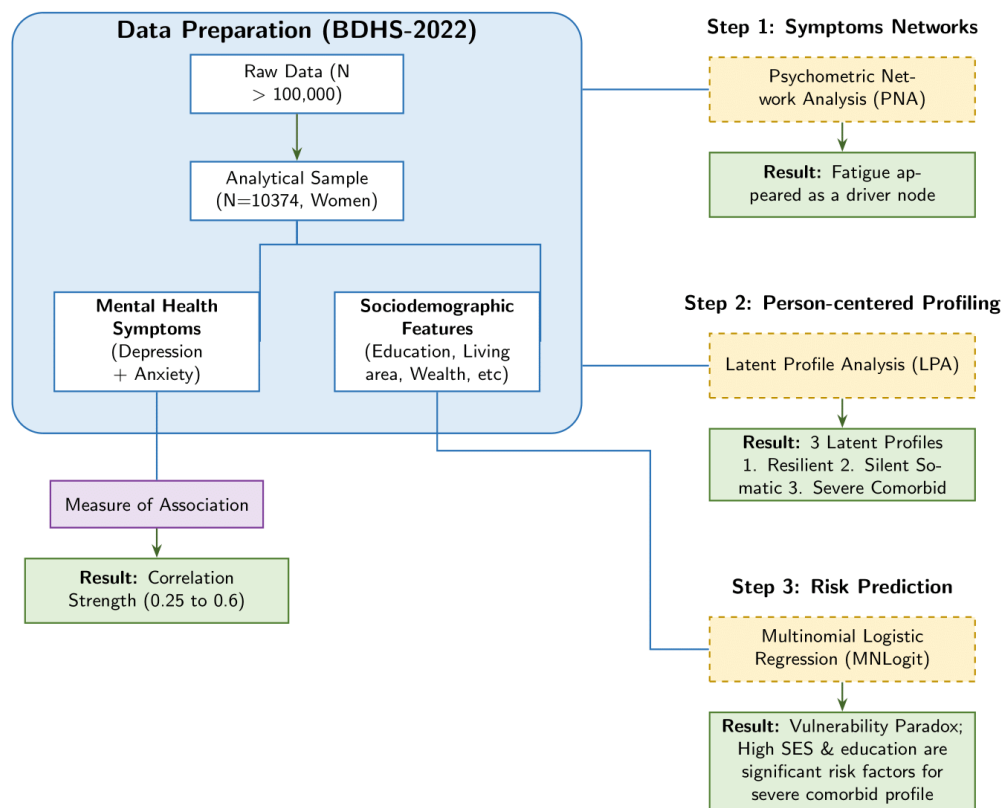


Figure 1: Workflow of the proposed network-informed latent profile analysis.

a nationwide cross-sectional survey of different age groups, mainly focused on social and health-related factors. Initially, the sample consisted of 5032 columns (variables) and more than 100,000 rows. However, the final 10,374 sample over 90 different variables of ever-married women of reproductive age was used for analysis.

3.2. Measures

The dataset can be broadly divided into two parts: the first one is mental health symptoms, and the other one is sociodemographic features.

3.2.1. Mental Health Symptoms

Patient Health Questionnaire (PHQ-9) and Generalized Anxiety Disorder (GAD-7) scale were the primary indicators for further analysis. The first 16 variables (labeled as mth1 to mth16), among which the first nine were depression, and the rest of them were anxiety-related. Responses taken as multiple-choice questions ranged from "Never" (0) to "Always" (3). Figure 2 shows the intercorrelation between the mental distress symptoms. The strength of correlation varies approximately between 0.25 to 0.55. The first five variables, such as apathy, sadness, sleep, fatigue, and appetite, labelled as (mth1 to mth5), showed relatively higher intercorrelations ($\rho \approx 0.40\text{--}0.55$), indicating potential core depressive or emotional symptoms and symptoms are related but not redundant means that symptoms are distinct but related aspects of mental health disorders. Moreover, no correlations exceed $\rho = 0.60$, indicating an absence of multicollinearity. This supports the suitability of these variables for multivariate modeling approaches such as latent profile analysis and psychometric network analysis. Table 1 shows some features with a few data points from mental health symptoms.

3.2.2. Sociodemographic Variables

These variables included a variety of associated socio-economic variables such as occupation, husband characteristics and behaviors, education level (no education to higher), religion, residence (urban/rural), and wealth index (poorest to richest). Table 2 shows a representative part of the sociodemographic variables.

3.3. Statistical Analysis

3.3.1. Psychometric Network Analysis (PNA)

PNA describes features (e.g., sadness, fatigue) conditional association, helping to understand symptoms entirely like a web rather than a cause of



Figure 2: Spearman correlation heatmap of mental health symptoms.

single symptoms . This study estimated the network structure of variables by the Gaussian Graphical Model (GGM) using the graphical lasso (glasso) method to identify the most influential (“driver”) symptoms. The nodes (circles) of the GGM model represent the features, and the edges (lines) represent the partial correlation of nodes [25]. Later on, this study applied a case-dropping bootstrap procedure with Spearman correlations to evaluate the Centrality stability. Figure 3 indicates that the centrality estimates remained above 0.95 even after dropping 70% samples, indicating highly stable centrality estimates.

The L1 regularization parameter α (alpha) was set to $\alpha = 0.01$ to reduce the regularization spurious association, overfitting, and boost the model stability since high-dimensional data was utilized. Diagonal elements of the precision matrix (inverse covariance matrix) were set to 0 to avoid self-loops. To measure the centrality indices (strength) of nodes, the sum of absolute

Variables	Obs	Mean	Std. Dev	Min	Max
Interest	10,374	0.7251	0.0756	0	3
Feeling down	10,374	0.3924	0.6279	0	3
Sleep	10,374	0.5531	0.6982	0	3
Feeling tired	10,374	0.3162	0.5918	0	3
Trouble relax	10,374	0.2043	0.5138	0	3
Restless	10,374	0.1933	0.4781	0	3
Irritable	10,374	0.0885	0.3544	0	3
Feeling afraid	10,374	0.0406	0.1974	0	3

Table 1: Representative part of mental symptoms data

Variables	Obs	Mean	Std. Dev.	Min	Max
Husband Edu	10,374	1.6507	1.0156	0	4
Worked 12mo	10,374	0.6360	0.9075	0	3
Religion	10,374	0.1161	0.3595	0	4
Residence	10,374	0.6351	0.5220	0	2
Smartphone	10,374	0.4554	0.4980	0	1
Internet use	10,374	0.3969	0.5027	0	2
Bank account	10,374	0.2530	0.4348	0	1
Wealth index	10,374	2.2689	1.3856	0	4

Table 2: Representative data from sociodemographic variables

partial correlation weights connected to each node is calculated. Features with the most centrality indices are interpreted as central features.

3.3.2. Latent Profile Analysis (LPA)

This study introduced LPA using Gaussian Mixture Modeling (GMM) to divide the mental health symptoms into several homogeneous subgroups. The number of subgroups is decided based on the Bayesian Information Criterion (BIC), Bootstrapped Likelihood Ratio Test (BLRT), and Entropy. It can be seen in Figure 4 and Table 3 that the value of BIC continued to decrease as more classes were added. Where the BIC value dropped noticeably for class three and diminished marginally afterwards. This study takes the BIC for class three as an elbow point, while the entropy value was also satisfying, more than 0.80, indicating the classes are well separated. Moreover, the

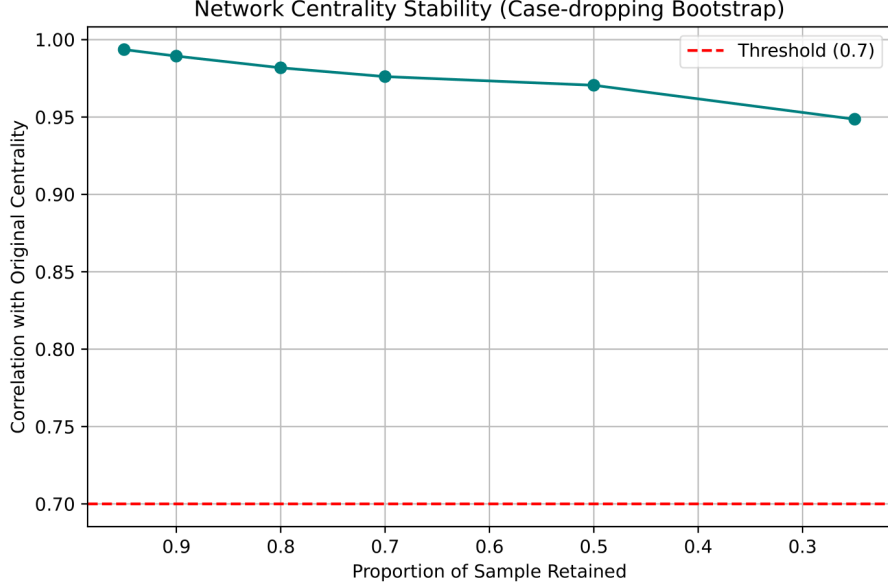


Figure 3: Measure of CS-coefficient of psychometric network analysis.

incremental gain of model fit from more than three profiles may not make up for the loss of parsimony and clinical interpretability. Therefore, due to the emergence of well-differentiated and theoretically meaningful profiles, interpretability, and parsimony, this study selected three profiles as an optimal solution [26]. The three classes (profiles) were labeled as (i) a low-symptom resilient class, (ii) a moderate “silent somatic” class, and (iii) a high-symptom severe comorbid class which are characterized as 25.6%, 49.4%, and 25.0% of the data, respectively. .

3.3.3. Multinomial Logistic Regression

Multinomial Logistic Regression (MNLogit) is an extended version of the logistic regression model in the case of predicting a nominal outcome variable with more than two categories. This study applied this model to several sociodemographic variables to estimate the odds ratios (OR) for different profiles created by LPA. Risk factors for the vulnerable groups were measured based on the Resilience Profile.

Classes	AIC	BIC	Log-likelihood	Entropy
2	217695.26	218397.93	-10.51	0.867
3	210005.87	211063.49	-10.14	0.835
4	206683.06	208095.63	-9.97	0.839
5	204660.42	206427.95	-9.87	0.794
6	203529.75	205652.23	-9.81	0.789

Table 3: Latent class analysis selection criteria

4. Results

4.1. Psychometric Network Analysis

The centrality measure of depression and anxiety symptoms is presented in Figure 5. It could be seen that “fatigue” comes up with the highest sum of partial correlation (1.958096), making fatigue a central node in the network. The second largest partial correlation is 1.713212 for feeling down or “sadness.” These indicate that fatigue and sadness may play a central role in overall mental disorders and work as the primary bridge in order to activate the wider symptom network in the population. Additionally, symptoms related to depression tend to affect mental health more than anxiety-related issues.

4.2. Latent Profile Analysis

Model fit indices supported a 3-profile solution (elbow point of BIC curve, significant BLRT ($p < 0.001$), Entropy > 0.80). The profiles, visualized in Figure 6, were characterized as follows:

- **Resilient Profile (25.6%):** Characterized by uniformly low symptom scores across all domains.
- **Silent Somatic Profile (49.4%):** A unique phenotype characterized by moderate scores on Fatigue and Sleep items but distinctly low scores on Fear and Panic (GAD-7 items).
- **Severe Comorbid Profile (25.0%):** The highest-risk group, exhibiting elevated scores across all 16 somatic, affective, and cognitive symptoms.

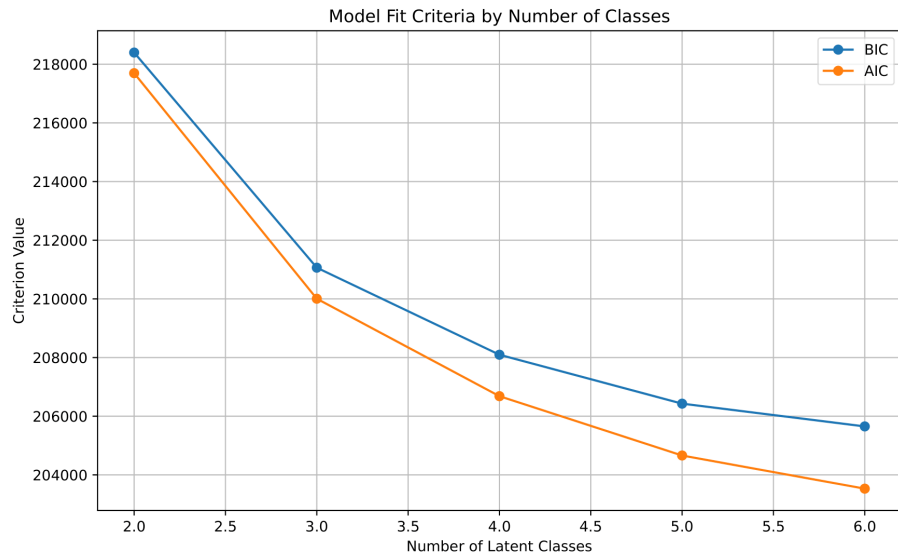


Figure 4: AIC & BIC curves for different latent classes.

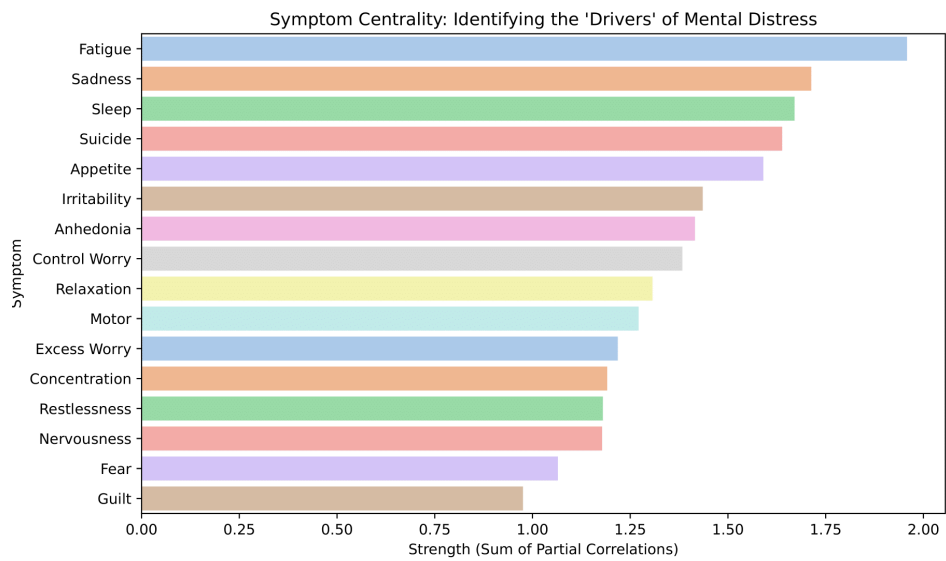


Figure 5: Identifying centrality by PNA.

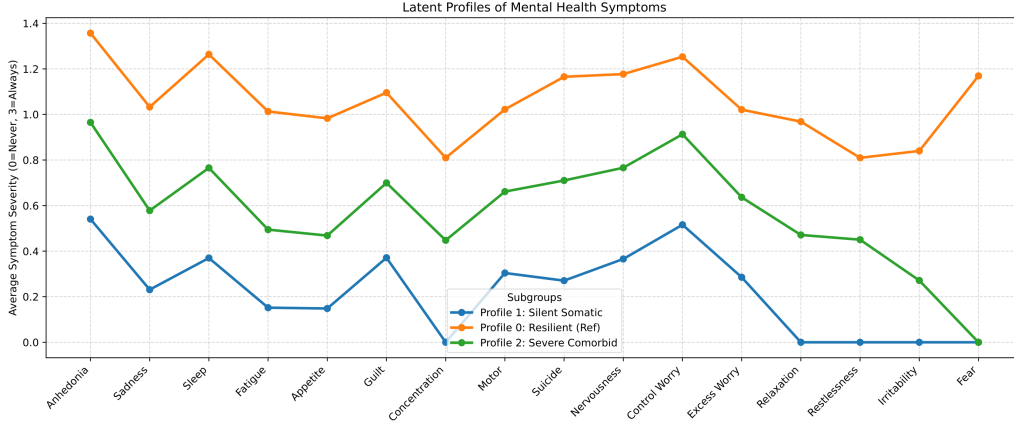


Figure 6: Visualization of three latent profiles.

4.3. Predictors of Profile Membership

The outcome of the Multinomial Logistic Regression (MNLogit) in Table 4 illustrates the predictors that appeared as statistically significant and paradoxical for the Severe Comorbid Profile with respect to the Resilient group.

In terms of education, this study finds higher education to be a significant risk factor. Women who completed secondary education had 73.36% lower odds of being classified in the severe comorbidity group compared to those with higher education, while it is 51.05% lower for primary education holders. Similarly, Urban residence and the Richest wealth status were associated with the highest likelihood of severe comorbidity. Regarding wealth status, among the groups (poorest, poorer, richer, middle), richer people have 70.01% lower odds of being in the severe group, while the middle class has 63.72% lower odds compared to the richest class. Rural residence shows 52.74% lower odds of falling into the severe group compared to urban residence.

5. Discussion

In Bangladesh, most of the studies are based on the regular statistical methods, whereas this study applies a network-informed latent profile approach to mental health. The findings of this study are not on correlation or association based which typically point out significant association of poverty, education, social media, and other features with potential health risk.

Predictor	Comparison	OR	95% CI	P-Value
Education	No Edu. vs. Higher	0.6549	[0.5045, 0.8500]	0.0015
	Primary vs. Higher	0.4895	[0.4019, 0.5961]	<0.001
	Secondary vs. Higher	0.2664	[0.2289, 0.3100]	<0.001
Wealth	Poorest vs. Richest	0.5163	[0.4152, 0.6421]	<0.001
	Poorer vs. Richest	0.4897	[0.4012, 0.5978]	<0.001
	Middle vs. Richest	0.3628	[0.2977, 0.4422]	<0.001
	Richer vs. Richest	0.2999	[0.2485, 0.3618]	<0.001
Residence	Rural vs. Urban	0.4726	[0.4168, 0.5359]	<0.001

Table 4: Multinomial Logistic Regression Predicting Latent Profile Membership

5.1. The Somatic-Energy Bridge

The crucial finding of the psychometric network analysis (PNA) is exhaustion, identified as the central node. In the PNA theory, the central node works as a bridge that influences the other symptoms [13]. This finding suggests that the mental health issues of Bangladeshi women result from physical depletion rather than cognitive issues. Since feeling bad about oneself or feeling guilt, nervousness & anxiety, and fear symptoms appeared with half strength compared with fatigue.

This is similar to the idea of the “Somatic Idioms of Distress” framework, where psychological distress is culturally expressed through physical depletion [13]. In Bangladesh, the idea of mind-body dualism in psychiatry is not as significant as in Western societies, meaning women tend to express their psychological imbalance through physical depletion rather than sadness or fear. According to social customs, admitting emotional weakness (e.g., “I am anxious or depressed”) is considered a lack of gratitude or spiritual beliefs. Conversely, expressing symptoms like fatigue, pain, and insomnia is widely supported in families, and family members also help with treatment. This physical treatment is generally based on physical symptoms without regarding mental condition, risking the large subgroups where pathology is encoded in their body. Moreover, a lack of proper knowledge about mental health symptoms or the consequences of symptoms may lead to biased answers. As a result, fatigue and insomnia appeared as central nodes in the PNA method, representing the cultural barriers to expressing mental distress.

5.2. *The “Silent Somatic” Phenotype*

The study labeled almost half of the samples (49.4%) of women as the “silent somatic” profile, who were struggling with irregular sleep and physical exhaustion but showed less affected by panic, fear, and worry, which are typical symptoms of anxiety diagnoses. Most of the methods used to diagnose anxiety are based on binary classification (yes or no); thus, women with a GAD-7 score of more than 10 ($score > 10$) may be classified in the anxiety category. However, since they lack panic and fear, they probably get a low score on the GAD-7 scale, which may lead to drawing a misleading conclusion about them. This explains why profile analysis is important for the study.

5.3. *The Paradox of Privilege*

The most striking finding of this study is the higher potentiality of the elite class, who possess higher education, the richest wealth status, and live in developed areas, of having mental distress issues compared to their counterparts. This paradox of women might be explained by the “double burden” term. While they are enrolling for higher education and entering the job market, they also need to maintain household activities like taking care of children, making dishes for family members, and following the customs and rules of society [27].

It would be more observable through the lenses of "Urbanization stress" and "Role Conflict Theory". The hypothesis of urbanization stress suggests that a shift in developed urban life often creates distance from the traditional joint-family support system. Highly educated, wealthy women often feel isolated and experience Relative Deprivation since they need to stay away from family or friends and live in urban facilities; thus, maintaining a high social status with traditional community support puts women in great psychological pressure.[28]

5.4. *Strengths and Limitations*

The main strengths of this study are the use of a nationally representative multidimensional (mental health symptoms and socio-demographic) dataset of the reproductive age. The PNA and LPA, the variable-center approach, figure out the central reasons for mental health disorders. This study points out that fatigue and insomnia contribute more than sadness and fear, the common reasons for mental issues. Additionally, it also finds that women living in urban areas with high social status are more exposed to mental disorders.

However, this work is not beyond limitations; there are several drawbacks that should be introduced, such as the use of cross-sectional and only women-based data from BDHS-2022. Although PNA identified fatigue as a driver node, this study failed to identify directions, like whether depression causes fatigue or fatigue causes depression. Then this study figured out the vulnerability paradox (Higher education, Higher risk), but the reasons behind this are not known. The dataset lacked specific indicators for work-life balance, academic pressure, and isolation, limiting this study’s ability to determine the reason for the double burden hypothesis. In the case of multinomial logistic regression, this study only accounts for three variables (education, living areas, and wealth status), while other variables can be used for a deeper analysis. Finally, in the Bangladeshi context, there is a greater probability that the data reported are biased since mental health issues are widely ignored by women in Bangladesh.

6. Conclusion

This study endeavors to do something different from the typical studies in Bangladesh. It reveals that fatigue and irregular sleep are key factors of women’s mental health complexity, which may influence the other associated factors. These two symptoms even have more influence on mental health than factors like worry, fear, and feeling guilty. Additionally, it reveals that women with a wealthy status and more educational qualifications tend to face mental issues. These findings may be explained through the lenses of the double burden theory, Urbanization Stress, and role conflict theory. The work-life imbalance, sticking to social customs, and taking care of family members while maintaining a job and pursuing higher education may finally show up as depression and anxiety. Therefore, it might be helpful for them if they take more rest, and sound sleep to alleviate mental health issues [29]. Moreover, their partners and family members should be supportive and come forward to help them with household chores and childcare to give them room for sufficient rest [30]. These findings also highlight the necessity of stratified public health interventions, which will focus on symptom pattern profiles and sociodemographic context to measure severity rather than solely relying on binary classification [29].

7. Declaration of interests

The authors declare no competing interests.

8. Data sharing statement

The data used in this study were obtained from the Demographic and Health Surveys (DHS) Program. The BDHS 2022 dataset is publicly available to registered users at <https://dhsprogram.com/>.

9. Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used Gemini (Google) in order to refine the clarity of the English text. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

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N/A

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